

Data Acquisition and Belief Formation*

Philipp Chapkovski¹, Arzu Işık², Mariana Khapko³, and Marius Zoican²

¹University of Duisburg-Essen

²Haskayne School of Business, University of Calgary

³Rotman School of Management, University of Toronto

Abstract

TBC

Keywords: experimental finance, market data, retail trading, order imbalance

JEL Codes: G11, G12, G14

*Philipp Chapkovski (chapkovski@gmail.com) is a postdoctoral researcher at the University of Duisburg-Essen. Arzu Işık is a doctoral student at Haskayne School of Business, University of Calgary. Mariana Khapko (mariana.khapko@rotman.utoronto.ca) is affiliated with University of Toronto Scarborough and the Rotman School of Management. Marius Zoican (contact author, marius.zoican@haskayne.ucalgary.ca) is the Canada Research Chair in Financial Technology at Haskayne School of Business, University of Calgary. We thank . . . for excellent research assistance. We are grateful to . . . for insightful discussions, as well as to seminar and conference participants at the Marius Zoican acknowledges funding from the Canada Research Chairs Program and the Social Sciences and Humanities Research Council of Canada (SSHRC Insight Grant 435-2025-0100).

1 Introduction

“Active retail investors in the U.S. equities market are demanding access to more granular information. Bring your investing decisions to the next level with depth-of-book data.” —

Nasdaq TotalView (webpage retrieved on February 12, 2025)

Retail investors increasingly pay for order flow data, yet little is known about how they value and use it. Online brokerage firms, exchanges, and retail-oriented financial platforms monetize market data through subscription-based analytics, premium order flow insights, and tiered access to real-time price feeds. For instance, Robinhood’s Gold subscription, priced at \$5 per month, provides users with Level II market data from Nasdaq, offering deeper insights into bid and ask prices. Interactive Brokers charges venue-dependent fees for non-professional traders, reaching up to \$135 per month for ICE Futures data. At the same time, TD Direct Investing charges up to \$32 per month for Level II data on U.S. equity markets, while in Canada, Questrade offers similar access for \$49.95 per month as of October 2025.

Retail trading has expanded dramatically over the past decade, reshaping market dynamics. The share of retail volume in U.S. equity markets surged from 10.1% in 2010 to over 23% by January 2023. Between January 2021 and August 2024, Robinhood’s assets under custody more than doubled, rising from \$63 billion to \$140 billion, underscoring the growing influence of retail investors in financial markets. Retail traders have also grown increasingly sophisticated, actively seeking out complex and speculative instruments: ? document that as of 2021, retail investors accounted for up to 60% of trading volume in U.S. options.

Can retail traders effectively learn from market data? Bartlett et al. (2023) and Kwan et al. (2020) leverage machine learning techniques to show that order flow data is highly valuable for algorithmic hedge funds. Professional traders invest heavily in infrastructure to extract short-term signals from order flow and trade on them within milliseconds, which gives them a measurable edge. Retail traders, by contrast, are not natural users of such data. They may lack the speed, automation, and analytical tools required to fully exploit order flow information. Why, then, do they pay for it? One possibility is that retail traders see access to order flow data as a mark of professionalism, aligning themselves with institutional practices even if they cannot replicate the same strategies. Another is that order flow data gives the illusion of control, making traders feel more informed, even if it does not materially improve their returns.

This paper provides the first controlled evidence on how retail traders value and use order-flow data when they must actively pay for it. We build on [Andries et al. \(2025\)](#) and develop an online randomized controlled experiment in oTree ([Chen et al., 2016](#)), where participants complete a sequence of independent investment rounds. In each round, they observe 16 historical realizations of returns and order-flow imbalance, mimicking a stylized intraday environment. Importantly, the imbalance is informative in only half of the rounds: in predictable rounds, lagged imbalance forecasts next-period returns with a correlation of roughly 0.67; in baseline rounds, the imbalance is pure noise. Participants are never told which rounds are predictable.

Our design cleanly separates two forces that drive the demand for data. First, we conjecture an *endowment effect*: the act of paying for data may distort beliefs ex post, altering both perceived informativeness and portfolio choices even though the underlying information is identical. Second, there is *selection effect*: participants vary in their willingness to pay depending on their financial literacy, overconfidence, and risk preferences. We implement two fee regimes to identify these channels. In the data-fee regime, participants may purchase access to the imbalance for a fixed fee each round; in the brokerage-fee condition, participants automatically receive the data for an equivalent fee. Because fees are symmetric across conditions, differences in behavior arise entirely from whether the participant made an *active* purchase.

2 Experiment design

2.1 Round investment game

Our experimental design builds on the framework of [Andries et al. \(2025\)](#). In each round, participants observe past realizations of two variables: (i) a “stock return” r_t and (ii) an order flow imbalance ω_t . The display consists of 16 past observations — visually mapped to half-hour intervals over a trading day — to mimic an intraday trading environment.

Across rounds, the order imbalance may or may not predict returns. Participants know this but are not told which specific rounds contain predictive information, nor the fraction of such rounds (set to 50%). We refer to rounds where the imbalance is uninformative as *baseline* rounds, and to those where it predicts returns as *predictable* rounds.

In baseline rounds, stock returns are drawn from a random walk process

$$r_{t+1}^b = \mu + \epsilon_{t+1}^b, \quad (1)$$

where $\mu = 15\%$ and $\epsilon_t^b \sim \mathcal{N}(0, \sigma_b^2)$ with $\sigma = 12\%$. Conversely, in predictable rounds, stock returns load on the lagged order imbalance:

$$r_{t+1}^p = \mu + \omega_t + \epsilon_{t+1}^p, \quad (2)$$

where the order imbalance is identically and independently distributed as $\omega_t \sim \mathcal{N}(0, \sigma_\omega^2)$ and $\epsilon_t^p \sim \mathcal{N}(0, \sigma_p^2)$. We impose $\sigma_p^2 + \sigma_\omega^2 = \sigma_b^2$ to ensure that r^b and r^p share the same unconditional distribution. The only difference between baseline and predictable rounds is thus the presence of an informative signal that is correlated with returns.

In the experiment, we set $\sigma_\omega = 8\%$ which implies $\sigma_p = 8.9\%$. This parametrization yields a correlation of $\text{Corr}(r_{t+1}^p, \omega_t) = \frac{0.08^2}{0.12 \times 0.08} = 0.67$, ensuring that the imbalance is sufficiently informative to be visually detectable in the time-series display, while still leaving substantial residual noise. This mirrors realistic intraday settings, where order-flow imbalance carries predictive content but is far from a perfect signal. Figure 1 illustrates the information displayed to participants in each round.

At the start of each round, participants receive an endowment of 100 E\$ (experimental dollars). They choose how much to invest in the risky stock; the remainder is held in cash at a zero return. Leverage and short selling are not permitted. In addition to choosing their portfolio, participants report (i) whether they believe the order-flow imbalance is informative in that round and (ii) their forecast of the next-period return.

At the end of the round, participants receive feedback on the realized return and on whether the imbalance was in fact informative. They earn E\$2 for correctly identifying the round type and an additional E\$2 for providing a forecast within one percentage point of the realized return.

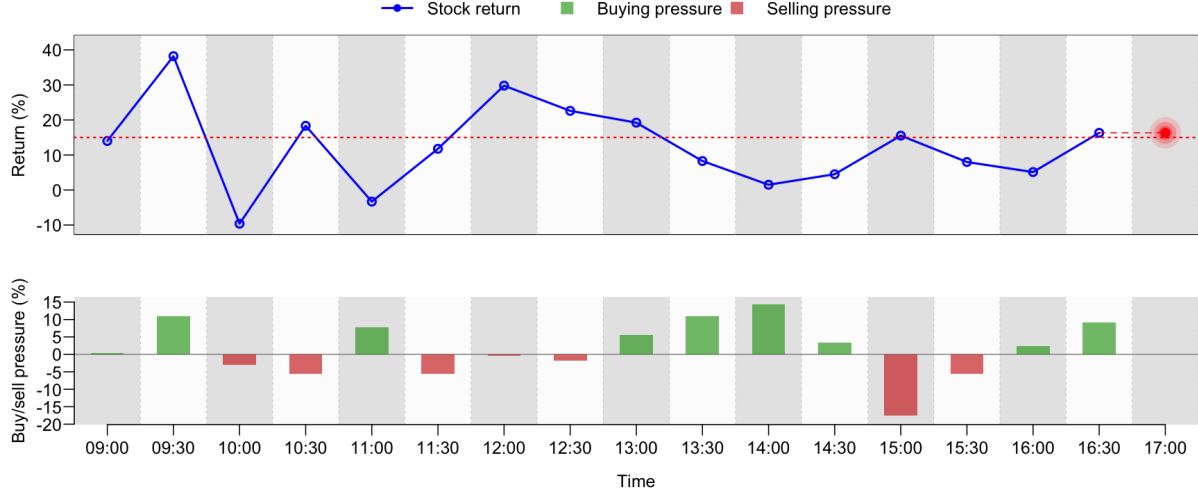
2.2 Estimating the value of order imbalance data

To quantify the ex ante value of data, we impose structure on investor preferences. We assume investors have constant relative risk aversion (CRRA) over wealth, $U(w) = \frac{w^{1-\gamma}}{1-\gamma}$, where γ is the risk aversion coefficient.

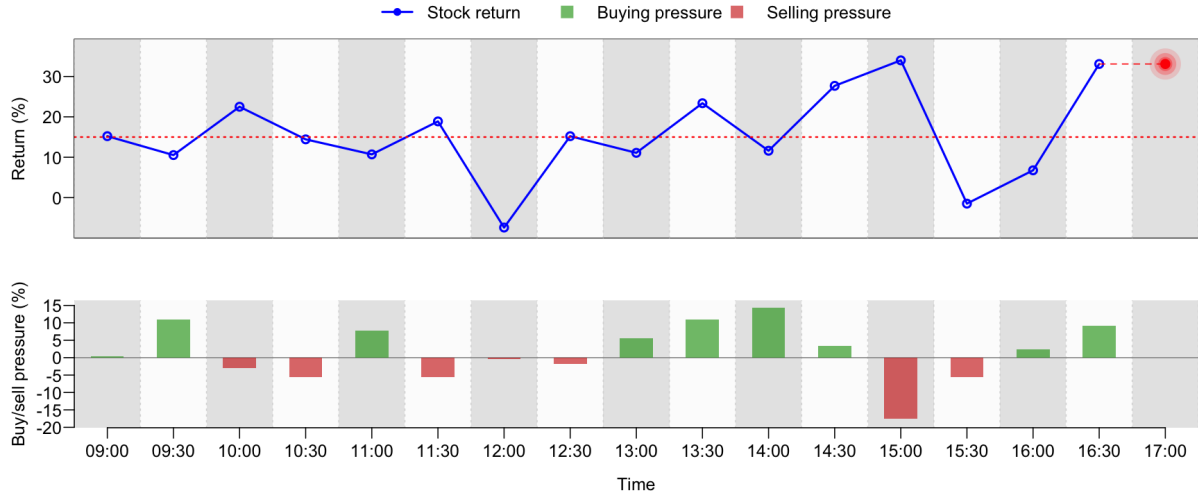
Under CRRA preferences and normally distributed returns, the classical result in Merton (1969) implies

Figure 1: **Example screens shown to participants.**

Each panel displays 16 past half-hour returns and order-flow imbalances (buy/sell pressure). The return to be predicted (at 17:00) is marked by a red fuzzy circle, and the unconditional mean return $\mu = 15\%$ is indicated by a dotted red line.



(a) Baseline round: order imbalance is uninformative.



(b) Predictable round: order imbalance is informative.

that investors allocate a constant share of wealth to the risky asset equal to $\alpha = \frac{\mathbb{E}r}{\gamma \sigma^2(r)}$. Using our experimental parameters and a risk-aversion level of $\gamma = 24$ (calibrated to estimates in [Andries et al., 2025](#)), the optimal portfolio share in baseline rounds is $\alpha_b^* = 43.4\%$. This allocation yields a certainty equivalent of E\$103.31 for an initial endowment of E\$100.

To estimate optimal investment in predictable rounds, we simulate 20,000 realizations of the order-flow imbalance. For each simulated draw ω_i , the optimal risky share is

$$\alpha_{r,i}^* = \max \left\{ 0, \min \left\{ \frac{\mu + \omega_i}{\gamma \sigma_p^2}, 1 \right\} \right\}, \quad (3)$$

that is, the Merton share projected onto the admissible interval $[0, 1]$ reflecting the no-leverage and no-shorting constraints.

Figure 2 illustrates the simulation results for $\gamma = 24$. Panel (A) plots the distribution of optimal risky shares $\alpha_{p,i}^*$, which vary substantially across simulations. The mean and median optimal shares are 70.6% and 78%, respectively: that is, an increase of roughly 62 percentage points relative to the baseline allocation. Predictive information therefore leads investors to take substantially larger positions on average, and it also generates considerable dispersion: in about one third of simulations, the optimal policy hits a constraint boundary (30% at the upper boundary of full investment, and 3% at the lower boundary of zero investment).

Panel (B) reports the corresponding distribution of certainty equivalents (CEs) across the 20,000 predictable-round simulations. The distribution is right-skewed: while most realizations yield moderate improvements over the baseline CE of E\$103.31, a meaningful subset delivers much larger utility gains when the order-flow signal strongly increases expected returns. The mean and median certainty equivalent in predictable rounds are E\$107.53 and E\$105.98, respectively, implying an ex-ante value of data of E\$4.22 (=E\$107.53-E\$103.31).

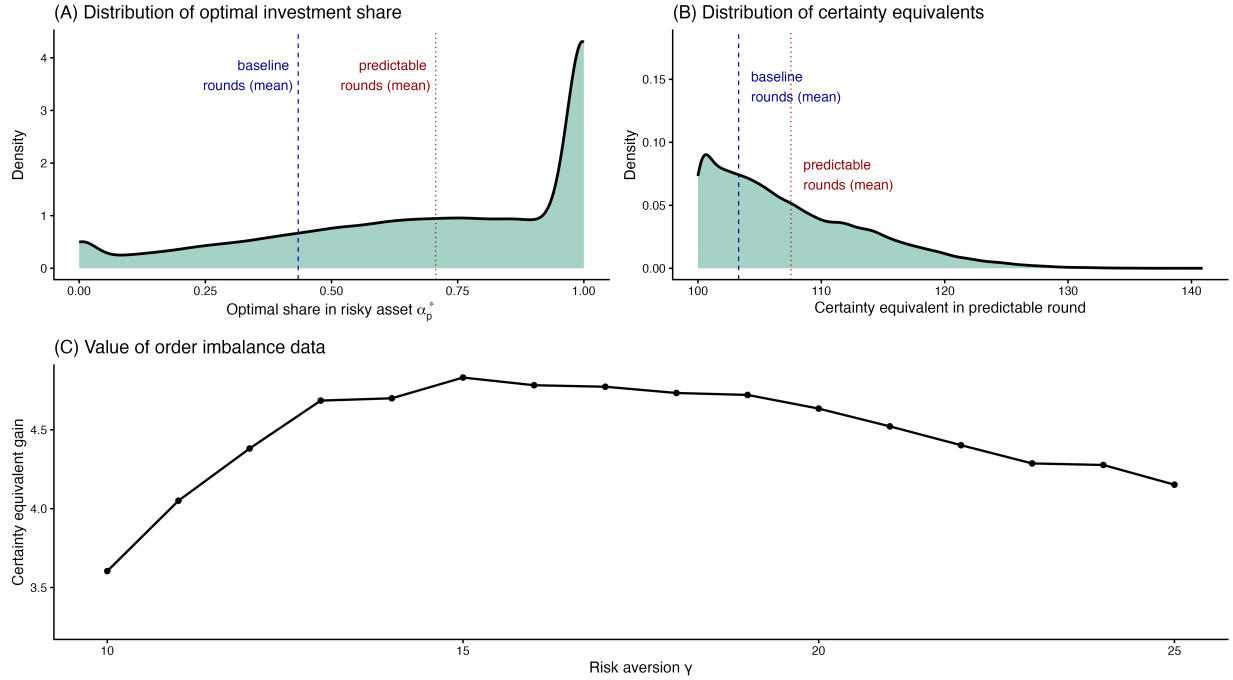
Panel (C) summarizes the ex ante value of order-flow imbalance data as a function of risk aversion γ . The certainty-equivalent gain is strictly positive for all γ considered, confirming that predictive order-flow information is always valuable, and it remains relatively stable around E\$4. On the one hand, data is valuable for risk-averse investors because it reduces the residual noise in future returns. On the other hand, as risk aversion increases, optimal positions shrink both with and without data, which dampens the incremental benefit of information.

2.3 Experiment design

All participants play 14 rounds and begin each round with an endowment of E\$100. The first two rounds are training rounds and always set data and brokerage fees to zero. In each subsequent round, participants face

Figure 2: Optimal portfolios and the value of predictive order flow data

This figure shows simulation results for the optimal risky share and certainty-equivalent gains under CRRA preferences. Panel (A) plots the distribution of optimal risky shares in predictable rounds. Panel (B) reports the corresponding distribution of certainty equivalents. Panel (C) shows the ex ante value of data as a function of risk aversion γ .



one of two conditions:

- **Data fee condition.** Participants may purchase order-imbalance data for E\$4 per round (the calibrated fair value from Section 2.2). Those who pay receive the data; those who decline see the message: “*You do not have privileges to access order-flow data.*” The brokerage fee is zero in this condition.
- **Brokerage fee condition.** Participants automatically pay a platform fee of E\$4 per round and always receive order-imbalance data. The data fee is zero in this condition.

Participants are randomly assigned to a **treatment** or **control** group. Control participants play all twelve non-training rounds under the brokerage fee condition. Treatment participants complete two four-round block in the data fee condition and one four-round block in the brokerage fee condition. The order of the three blocks is randomized at the participant level to mitigate learning or sequencing effects. Within each block, order-imbalance data is informative in exactly half of the rounds.

Finally, participants answer 13 financial literacy questions provided in Appendix A. The questions are

adapted from [Fernandes et al. \(2014\)](#) and include the three-question measure developed by [Lusardi and Mitchell \(2011\)](#), as well as two trading specific questions following the MiFID questionnaire in [Inghelbrecht and Tedde \(2024\)](#). To measure overconfidence and distinguish between subjective and objective financial literacy, we also ask participants:

On a scale from zero to ten, where zero is not at all knowledgeable about personal finance and ten is very knowledgeable about personal finance, what number would you be on the scale?

This question measures self-assessed (subjective) financial knowledge and is identical to the one used in [Cupák et al. \(2020\)](#). We pose this question before the experiment begins to avoid any influence from the trading game’s monetary performance or the quiz’s perceived difficulty.

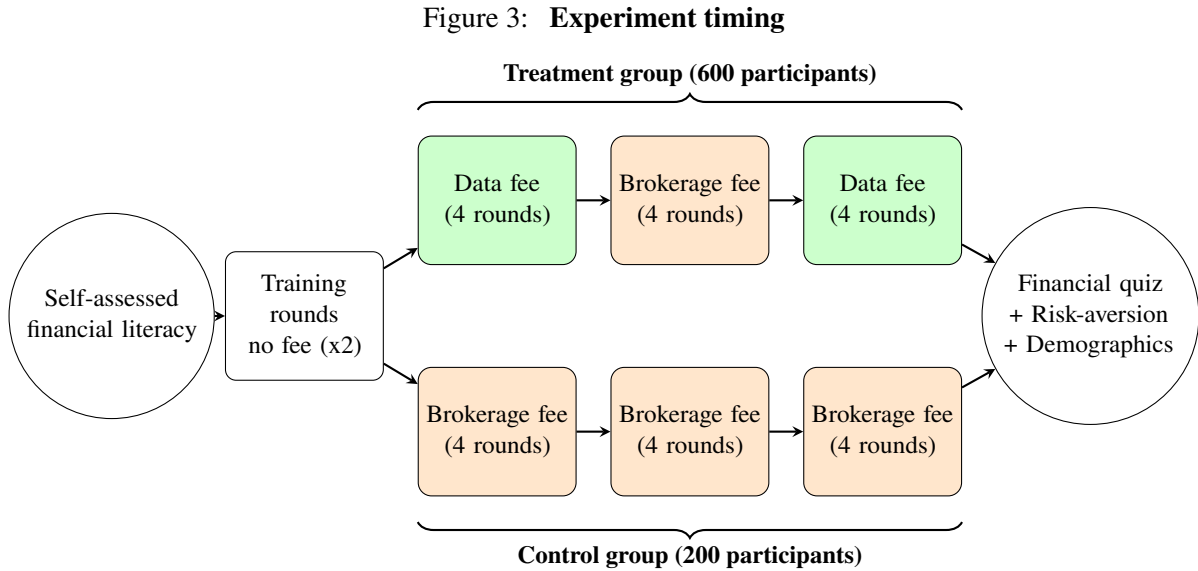


Figure 3 illustrates the experiment’s timeline. Appendix B reproduces the consent form. The experimental instructions given to participants are reproduced in Appendix C. Before the trading starts, participants need to correctly answer six comprehension questions, listed in Appendix D. This allows us to make sure that participants indeed understand the experiment before the trading rounds start. Finally, at the end of the experiment all participants are required to fill in a risk-aversion ([Holt and Laury, 2002](#)) and a demographic questionnaire.

3 Hypotheses

Our framework isolates two distinct forces driving the demand for order-flow data:

- (i) *Endowment effect.* Actively paying for data distorts beliefs and decisions *ex post*, holding the information set fixed.
- (ii) *Selection effect.* Participants differ in their *ex ante* willingness to pay for data depending on their financial literacy, overconfidence, and risk-aversion.

Since participants in the treatment and control groups receive identical information in any given round, any differences in beliefs or investment behavior between the treatment and control groups must arise from the act of purchasing the data rather than from differential information sets.

Hypothesis 1. *Belief endowment effect.* *Participants who pay for order-flow data perceive the imbalance as informative more often than participants who receive identical data for free.*

The intuition is straightforward. Once participants pay the data fee, they become motivated to believe the expenditure was worthwhile. The purchased signal is mentally coded as an “asset” they now own, which inflates its perceived value and predictive content. As a result, participants who pay are more likely to interpret noise as information and to assign higher subjective informativeness to identical order-flow patterns.

Under standard CRRA preferences with normally distributed returns, the optimal risky share is invariant to fixed fees (Merton, 1969). Thus, paying for data cannot change optimal investment through a classical expected-utility channel. However, if paying increases the subjective informativeness of the signal, then both forecasts and portfolio choices will respond more strongly to movements in the order-flow imbalance. In this sense, payment magnifies the behavioral impact of the data: it raises the elasticity of both beliefs and risk-taking with respect to the underlying signal, even though the signal itself is identical.

Hypothesis 2. *Investment endowment effect.* *Conditional on having access to order-flow data, participants who pay for data adjust their risky investment more strongly in response to order flow imbalance than participants who receive identical data for free.*

Hypothesis 3. *Forecast endowment effect.* *Conditional on having access to order-flow data, participants who pay for data adjust their return forecast more strongly in response to order flow imbalance than participants who receive identical data for free.*

We further conjecture that paying for information raises the elasticity of investment with respect to beliefs by making data more salient in the portfolio choice process: that is, paying for data improves the pass-through from forecasts to investment choices.

Hypothesis 4. *Belief pass-through.* *Paying for order-flow data increases the sensitivity of portfolio choices to participants' own return forecasts. Investors who pay for data place greater weight on their beliefs when allocating wealth to the risky asset.*

Participants who overestimate their financial ability believe they can extract more information from the signal, leading them to rely more heavily on data once they have “bought in.”

Hypothesis 5. *Overconfidence and beliefs.* *Overconfident participants exhibit stronger belief and investment endowment effects: paying for data increases both perceived informativeness and investment more for overconfident individuals.*

Next, we conjecture that the degree of financial literacy disciplines expectations. That is, participants with stronger financial knowledge are less susceptible to sunk-cost justifications.

Hypothesis 6. *Financial literacy and beliefs.* *Participants with higher financial literacy exhibit weaker endowment effects.*

The decision to purchase data reflects heterogeneous ex-ante valuations. Overconfident individuals believe they can convert information into profitable trades, leading to higher willingness to pay. Financially educated participants may value the data for more rational reasons: they better understand how order-flow signals can, in principle, improve forecasts.

Hypothesis 7. *Selection effect.* *Overconfident participants and participants with formal financial education are more likely to purchase data in the data-fee condition.*

The value of information is non-linear in risk aversion. Very risk-tolerant participants already invest aggressively, so incremental information has limited marginal benefit. At the other extreme, highly risk-averse

participants invest conservatively even with perfect information, reducing their willingness to pay. The predicted pattern is therefore hump-shaped: intermediate levels of risk aversion generate the highest private value of predictive data.

Hypothesis 8. *Risk aversion.* *Willingness to pay for data is lowest among the most risk-averse and the least risk-averse participants, producing a non-monotonic relationship.*

In baseline rounds, the order-flow imbalance carries no predictive content and therefore a Bayesian agent should rationally ignore it. However, participants who pay for the data are motivated to use it, and they interpret random fluctuations as meaningful patterns. This effect produces systematic overfitting and forecasts become noisier.

Hypothesis 9. *Noise overfitting.* *In baseline rounds, where the imbalance is truly uninformative, paying for data worsens forecast accuracy as participants overfit noise.*

Financially literate participants are better at incorporating the imbalance into their forecasts. As a result, the same data yields greater predictive gains for them and improves forecast accuracy.

Hypothesis 10. *Learning from data.* *In predictable rounds, order flow data improves forecast accuracy particularly for participants with high financial literacy.*

References

- Andries, M., Bianchi, M., Huynh, K. K., Pouget, S., 2025. Return predictability, expectations, and investment: Experimental evidence. *The Review of Financial Studies* 38, 1687–1729.
- Bartlett, R. P., McCrary, J., O’Hara, M., 2023. The market inside the market: Odd-lot quotes. *The Review of Financial Studies* 38, 661–711.
- Chen, D. L., Schonger, M., Wickens, C., 2016. oTree—an open-source platform for laboratory, online, and field experiments. *Journal of Behavioral and Experimental Finance* 9, 88–97.
- Cupák, A., Fessler, P., Hsu, J. W., Paradowski, P. R., 2020. Confidence, financial literacy and investment in risky assets: Evidence from the survey of consumer finances. *Finance and Economics Discussion Series* 2020.0.

- Fernandes, D., Lynch, J. G., Netemeyer, R. G., 2014. Financial literacy, financial education, and downstream financial behaviors. *Management Science* 60, 1861–1883.
- Holt, C. A., Laury, S. K., 2002. Risk aversion and incentive effects. *American Economic Review* 92, 1644–1655.
- Inghelbrecht, K., Tedde, M., 2024. Overconfidence, financial literacy and excessive trading. *Journal of Economic Behavior & Organization* 219, 152–195.
- Kwan, A., Philip, R., Shkilko, A., 2020. The conduits of price discovery: A machine learning approach. *SSRN Electronic Journal* .
- Lusardi, A., Mitchell, O., 2011. Financial literacy around the world: An overview .
- Merton, R. C., 1969. Lifetime portfolio selection under uncertainty: The continuous-time case on JSTOR. *Review of Economics and Statistics* 51, 247–257.

Online Appendix

December 9, 2025

A Financial literacy quiz

1. Suppose you had \$100 in a savings account and the interest rate was 3% per year. After 4 years, how much do you think you would have in the account if you left the money to grow?
 - (a) More than \$112
 - (b) Exactly \$112
 - (c) Less than \$112
 - (d) Don't know / Not sure
 - (e) Prefer not to say
2. Imagine that the interest rate on your savings account was 1% per year and inflation was 2% per year. After 1 year, with the money in this account, would you be able to buy
 - (a) More than today
 - (b) Exactly the same as today
 - (c) Less than today
 - (d) Don't know / Not sure
 - (e) Prefer not to say
3. A 15-year mortgage typically requires higher monthly payments than a 30-year mortgage, but the total interest paid over the life of the loan will be less.
 - (a) True
 - (b) False
 - (c) Don't know / Not sure
 - (d) Prefer not to say
4. Normally, which asset described below displays the highest fluctuations over time?
 - (a) Savings accounts
 - (b) Stocks
 - (c) Bonds
 - (d) Don't know / Not sure
 - (e) Prefer not to say
5. When an investor spreads his money among different assets, does the risk of losing a lot of money:
 - (a) Increase

- (b) Decrease
 - (c) Stay the same
 - (d) Don't know / Not sure
 - (e) Prefer not to say
6. Considering a long time period (for example, 10 or 20 years), which asset described below normally gives the highest return?
- (a) Savings accounts
 - (b) Stocks
 - (c) Bonds
 - (d) Don't know / Not sure
 - (e) Prefer not to say
7. Do you think that the following statement is true or false? "If you were to invest \$1,000 in a stock mutual fund, it would be possible to have less than \$1,000 when you withdraw your money."
- (a) True
 - (b) False
 - (c) Don't know / Not sure
 - (d) Prefer not to say
- (a) The difference between the lowest ask price and the highest bid price.
 - (b) The average of the bid price and the ask price.
 - (c) The fee the broker charges each time you trade.
 - (d) The change in the stock price from market open to market close.
 - (e) Don't know / Not sure
 - (f) Prefer not to say
8. Which of the following statements is correct?
- (a) Once one invests in a mutual fund, one cannot withdraw the money in the first year
 - (b) Mutual funds can invest in several assets, for example invest in both stocks and bonds
 - (c) Mutual funds pay a guaranteed rate of return which depends on their past performance
 - (d) None of the above
 - (e) Don't know / Not sure

- (f) Prefer not to say
9. Which of the following statements is correct? If somebody buys a bond of firm B:
- (a) She owns a part of firm B
 - (b) She has lent money to firm B
 - (c) She is liable for the company's debts
 - (d) None of the above
 - (e) Don't know / Not sure
 - (f) Prefer not to say
10. Suppose you owe \$3,000 on your credit card. You pay a minimum payment of \$30 each month. At an annual percentage rate of 12% (or 1% per month), how many years would it take to eliminate your credit card debt if you made no additional new charges?
- (a) Less than 5 years
 - (b) Between 5 and 10 years
 - (c) Between 10 and 15 years
 - (d) Never
 - (e) None of the above
 - (f) Don't know / Not sure
 - (g) Prefer not to say
11. When placing an order, you may see several prices: the last traded price, the ask price, the bid price, and the closing price. What is the bid price?
- (a) The highest price a buyer is currently willing to pay.
 - (b) The price you personally choose when submitting a buy order.
 - (c) The lowest price a seller is currently willing to accept.
 - (d) The most recent price at which the security traded.
 - (e) Don't know / Not sure
 - (f) Prefer not to say
12. When viewing a security's quote, you may see both a bid price and an ask price. What is the bid-ask spread?

B Consent form

Dear Participant,

You are invited to take part in a research study on financial decision-making funded by the Social Sciences and Humanities Research Council of Canada. The purpose of this study is to understand how people use data when making investment choices.

Your participation in this study is entirely voluntary. Participation presents no known risks or burdens. You will receive a compensation for participating in the study, announced on the Prolific platform, as well as a monetary compensation proportional to your performance in the experiment. You can withdraw from the study at any time without penalty. However, if you withdraw before the end of the experiment the compensation will be forfeit, as per Prolific's rules.

Please be assured that all data generated during this study will remain confidential. You will be asked to provide some basic demographic information about yourself, but none of this information will in any way be used to personally identify you and no personal information will appear in any published study.

Your decision to participate in this study will remain completely confidential and only the researchers will have access to any of the individual data. We do not collect any data that can personally identify you. Any demographics data will be stored electronically on a web-based server until it is downloaded for the purpose of analysis. This server will only be accessible by the research team using a secure password. Data will be retained in de-identifiable format for 7 years and discarded afterwards. At that time, electronic files will be deleted and written files will be shredded. This is a standard practice for the American Psychological Association. As required by the guidelines of American Economic Association, during the publication process the data might be shared with other investigators for re-analysis. However, to ensure security and confidentiality of the data files only details of the computations sufficient to permit replication might be shared.

This study has been reviewed by the Research Ethics Boards at the University of Calgary. If you are interested, you can receive a summary of the research study by emailing the researchers.

The research study you are participating in may be reviewed for quality assurance to make sure that the required laws and guidelines are followed. If chosen, (a) representative(s) of the Human Research Ethics Program (HREP) may access study-related data and/or consent materials as part of the review. All information accessed by the HREP will be upheld to the same level of confidentiality that has been stated by the research team.

If you wish to obtain further information regarding your rights as a participant you may contact the Conjoint Faculties Research Ethics Board (CFREB) at University of Calgary at cfreb@ucalgary.ca or (403) 220-8640. You may also contact the researchers for further information or to obtain answers to any questions about this research.

I agree to participate in research being conducted by the researchers.

I understand that I can withdraw my consent to participate at any time and by giving consent I am not giving up any of my legal rights.

I have read and understood the information above.

C Instructions

Welcome to this trading experiment! Please read these instructions carefully.

Overview. In each round, you will see a graph showing past data for a traded stock, measured over half-hour intervals within a single trading day. The graph displays:

1. **The stock return**, expressed in percentage terms; and
2. **The buy/sell imbalance**, a measure of order flow that captures buying versus selling pressure, expressed in percentage terms. It is positive when buy order flow exceeds sell order flow, and negative when sell order flow exceeds buy order flow.

In some rounds, the buy/sell imbalance is informative and helps predict the next stock return. In other rounds, the two variables are independent, and the imbalance does *not* help predict returns.

Your task. In each round, you are endowed with 100 experimental dollars (E\$). Your tasks are:

1. In some rounds, decide whether you want to purchase access to the buy/sell imbalance data.
2. Assess whether the buy/sell imbalance can be used to predict the next stock return.
3. Provide a forecast for the next stock return.
4. Choose how much of your endowment you would like to invest in the stock.

There are twelve main rounds in the experiment. All rounds are independent, and the average stock return is 15%. Before the main rounds begin, you will complete two practice rounds to familiarize yourself with the interface.

After each round, you will see the realized stock return, whether the buy/sell order flow imbalance was actually informative, the accuracy of your forecast, and the total wealth you earned in that round.

Your payoff. Your final payoff comprises of four parts:

1. The value of your portfolio (stocks plus cash) minus any brokerage or data access fees, from one randomly selected round.
2. E\$1 for every correct assessment of whether the order-flow imbalance can predict returns.
3. E\$1 for every precise forecast (within $\pm 1\%$ of the actual stock return).

4. E\$0.25 for each correct answer in a financial literacy quiz which you will complete after the main rounds.

At the end of the experiment, one of the twelve main rounds will be randomly selected to determine your investment payoff (component 1). Your total payoff in ECU is the sum of (1) from the selected round, plus (2) and (3) accumulated across all rounds, plus (4) from the financial literacy quiz. Your final payment in EUR is your payoff in ECU divided by 20.

D Comprehension quiz

1. Is the buy/sell order flow imbalance always useful for predicting the next stock return?
 - (a) Yes, it is always informative.
 - (b) No, it is never informative.
 - (c) It is informative in some rounds and uninformative in others.
 - (d) It depends on your investment choice.
2. What happens to the portion of your endowment that you *do not* invest in the stock?
 - (a) It earns interest.
 - (b) It is lost.
 - (c) It remains in cash and stays unchanged.
 - (d) It is automatically invested for you.
3. When is your stock forecast rewarded as precise?
 - (a) When it is exactly equal to the realized stock return.
 - (b) When it is within $\pm 1\%$ of the realized stock return.
 - (c) When it is within $\pm 5\%$ of the realized stock return.
4. Your investment choices matter because:
 - (a) You will be paid only if you have an average positive return across all rounds.
 - (b) You will receive your investment profit in every round in addition to your participation fee.
 - (c) You will receive your investment profit from *one randomly chosen round* in addition to your participation fee.